Despeckling SAR Images With Log-Yeo–Johnson Transformation and Conditional Diffusion Models

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***Abstract*— Satellite images of synthetic aperture radar (SAR) sensors are contaminated by speckles from the coherent imaging mechanism. Although removing or mitigating speckle has been a critical issue for SAR applications, effective reduction continues to be a significant challenge for existing methods when preserving the intricate structures within SAR images. To address this issue, this work proposes a novel conditional diffusion model for SAR despeckling (DiffusionSAR). The new method explicitly learns data distributions by forward diffusion toward multiplicative gamma noise. The logarithmic and Yeo–Johnson (log-Yeo–Johnson) transformation are harnessed in preprocessing for fine-tuning or hybrid training. A prolonging steps technique is suggested in fine-tuning to match the preprocessing. A new synthetic dataset is designed for satellite SAR despeckling. The proposed method is compared with eight state-of-the-art methods using both synthetic and real-world SAR satellite images. The qualitative and quantitative evaluations confirm the effectiveness of the proposed method in structural preservation as well as noise reduction. A fine-tuning experiment using stacked multitemporal data shows the necessity of tine-tuning training in bridging the domain gap when trained with synthetic data and tested with real-world SAR data.**

***Index Terms*— Despeckling, diffusion model, fine-tune, Sentinel-1, Sentinel-2, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a coherent imaging technology with strong invariance across diverse environmental conditions. This outstanding attribute makes SAR widespread in a multitude of fields, ranging from landscape classification to disaster monitoring, and surface change detection [1]. SAR images are, however, adversely affected by speckle, leading to a degradation in their quality. Speckle is observed in the granular or grainy appearance in SAR imagery. It is caused by the interference of the radar waves with each other when they bounce off the target and return to the radar sensor. The speckle pattern obscures image details and poses challenges for various applications and information extraction techniques. Therefore, removing speckle from SAR images is critical before they are put to practice.

Many despeckling methods have been proposed for SAR images, which can be categorized into local, nonlocal, and deep-learning-based methods. Local methods smooth local features with bilateral filters [2] or total variation (TV) regularization [3], [4] for homogeneous areas. Nonlocal methods [5], [6], [7], [8], [9], [10] are weight-based using nonlocal similarity between patches for matching. Deep-learning-based methods use convolutional neural networks (CNNs) [11], [12], [13], generative adversarial network (GAN) [14], [15], attention mechanism [16], transformer [17], and diffusion models [18], [19]. Unsupervised despeckling technologies [20], [21], [22], [23], [24] were also investigated to account for the absence of clean images.

The diffusion models for SAR despeckling [18], [19] have initially validated the effectiveness of diffusion models for SAR despeckling, but they are neither comprehensive nor accurate to explain the effectiveness of diffusion models for SAR speckling. Intuitively, the speckles are usually considered as the multiplicative gamma noise, in which the reason that the standard denoising diffusion probabilistic model (DDPM) can work well was not uncovered in [18]. The logarithmic transformation was suggested in [19], but the transformed gamma noise does not follow the standard Gaussian distribution well.

In this article, a comprehensive study is proposed to challenge the SAR image despeckling issue with diffusion models. A new conditional diffusion probabilistic model is proposed. The model is similar to the existing work, but the loss functions are enhanced for better structure preservation. New preprocessing and postprocessing schemes are proposed with the logarithmic and Yeo–Johnson (log-Yeo–Johnson) transformation for fine-tuning or hybrid training that aims to shorten the domain gap between optical and SAR images. To match the resolution of satellite SAR images, a new training dataset is constructed with 2400 pieces of Sentinel-2 image patches covering four main scenes. Besides the conventional test, a fine-tuning strategy is suggested to match the proposed diffusion models for better tradeoff between smoothness and

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strong scatters. The proposed method is named as the diffusion model for SAR despeckling (DiffusionSAR), and abbreviated as DiffusionSAR for convenience.

II. Related Work and Diffusion Models

A real-world SAR image y is usually considered as a clean image x_0 contaminated by the speckle which is typically modeled as multiplicative noise [25] as

$$y = x_0 \cdot n \tag{1}$$

where n denotes the speckle following the gamma distribution. According to [25], the probability density function (pdf) of the noise n is defined with the unit mean value and the $1/L$ variance as

$$p(n) = \frac{1}{\Gamma(n)} L^L n^{L-1} e^{-Ln} \tag{2}$$

where $\Gamma(\cdot)$ denotes the gamma function and L denotes the number of equivalent looks in the SAR image.

A. Related Work

Local methods were first proposed for SAR image despeckling aiming to reduce speckle by averaging pixels within a homogeneous region. For example, Wang et al. [2] proposed an SAR-oriented bilateral filter to explore local features. Adaptive spatial filters run a moving window for smoothness. The constraint of minimizing the TV was also used to regularize the recovered images, where despeckling is considered as a variational problem to obtain clean images by minimizing energy functions. For example, Aubert and Aujol [3] proposed a variational method in the original intensity domain based on the maximum a posteriori, and Zhao et al. [4] proposed an adaptive TV regularization to estimate the noise level of the SAR image with the wavelet decomposition and adapt the regularization parameter accordingly.

Nonlocal methods assign weights for similar pixels based on the similarity to the reference pixel. The similarity is typically measured as a decreasing function of the weighted Euclidean distance. A contribution to this denoising framework is the nonlocal-means (NL-means) algorithm proposed by Buades et al. [5]. Following this framework, Deledalle et al. [6] extended NL-means by introducing a probabilistic patch-based (PPB) filter. Parrilli et al. [7] introduced the 3-D block-matching (BM3D) algorithm for SAR denoising with the SAR-BM3D algorithm. Additionally, fast adaptive nonlocal SAR (FANS) [8], nonlocal anisotropic despeckling (NOLAND) [9], and model-free denoising algorithm for SAR images (MEET-SAR) [10] were proposed with nonlocal filters for SAR denoising. These methods require a considerable amount of time to process immensely sized images and their performance is highly dependent on the setting of filtering parameters and definition of similarity criterion.

Compared to traditional denoising methods that harness only the single input image, deep-learning-based methods have shown promising results by learning various features from rich datasets. Chierchia et al. [11] suggested the CNN for SAR image despeckling. Since neural networks for denoising usually deal with additive noise, the homomorphic approach with the logarithmic and exponential transformation were used in [11] to transfer the multiplicative noise to additive noise when feeding the SAR data into networks. The encoding–decoding framework was used in [12] to train an end-to-end network for SAR image despeckling. The multiobjective network (MONet) [13] employed a multiobjective loss function for CNN training. GAN was explored in [14] and [15] for SAR image despeckling. The attention mechanism [16] and transformer [17] have also been used for SAR image despeckling. These methods emphasized the importance of learning huge amount of data for inferring clean SAR images. The attention mechanism provides more powerful generalization ability than CNNs. Recently, the diffusion probabilistic model [26] has shown outstanding performance in the field of content generation. Dhariwal and Nichol [27] demonstrated that the diffusion model can achieve better quality than all other generative models in detail production. A large amount of research has proved that images follow a roughly uniform distribution (e.g., the Gaussian mixture model), but each image follows a specific distribution, too. CNNs and Transformers approximates the uniform model by shortening the gap between training targets and outcomes. In contrast, the diffusion model explicitly learns this distribution by feeding the targets as the input of neural networks. Researchers have proposed diffusion solutions for SAR despeckling. Perera et al. [18] suggested the DDPM directly for SAR image despeckling. Xiao et al. [19] despeckled SAR images with DDPM in the logarithmic domain to approximate gamma noise.

Since speckle-free SAR images are missing, unsupervised despeckling technologies were studied to avoid possible preparation of clean images. Boulch et al. [20] proposed a self-supervised SAR despeckling method based on the training of multitemporal real SAR images, where despeckling was achieved by training a neural network to produce an output as close as possible to another image representing the same scene at a different moment. However, they assumed that the multitemporal SAR images used for training are different only due to scattering noise, which is unreasonable because features often change according to climate or human factors. To address this problem, Ma et al. [21] proposed a noise-referenced SAR deep learning filter to optimize the training process using the statistical properties of scatterers to evaluate the similarity within each pixel pair between different images, which can mitigate the training error due to land cover changes. A refinement step was utilized in [22] to obtain compensation for changes. Molini et al. [23] proposed a self-supervised Bayesian denoising method based on blind-spot convolution. This method assumes that speckles are spatially uncorrelated and can be trained directly on real SAR datasets without the need for clean images. Additionally, Dalsasso et al. [24] proposed a self-supervised strategy based on the separation of the real and imaginary parts, which adequately considers the spatial correlation caused by the SAR transfer function specific to a given sensor and imaging mode.

In addition to unsupervised methods, multitemporal fusion and synthetic datasets are also suggested for supervised learning. Multitemporal fusion constructs a reference image from

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a multitemporal real-world SAR image [11] as approximately clean. Synthetic datasets take optical images as a clean source and inject simulated speckles for training [14]. Lattari et al. [28] proposed a two-phase learning strategy by pretraining the model and then fine-tuning the pretrained model using multitemporal phase fusion. Dalsasso et al. [22] adopted a similar approach by performing self-supervised pretraining on synthetic datasets and then self-supervised fine-tuning training on noisy image pairs. In terms of data size, it is difficult to obtain a large amount of training data through multitemporal fusion, but synthetic datasets are far larger. However, real-world SAR images are characterized by abrupt reflectivity changes, geometric distortions, and high dynamics, which cannot be fully simulated by synthetic datasets.

B. Diffusion Models for SAR Despeckling

A diffusion model for SAR image despeckling is based on the conditional diffusion model. As shown in Fig. 1, a standard diffusion model consists of a forward diffusion process and a backward denoising process.

1) Forward Diffusion Process: The forward process consists solely of adding noise to the image. It starts with a reference image x_0 derived from optical data. This reference image undergoes a series of iterations where simulated noise is gradually added at each step, forming a sequence of noisy images x_t. Each noisy image x_t is typically obtained by adding random noise to the previous image x_{t-1}. The transfer probability $q(x_t|x_{t-1})$ from x_{t-1} to x_t is a Gaussian distribution with the mean value $\mu_t = (1-\beta_t)^{1/2}x_{t-1}$ and the variance $\sigma_t^2 = \beta_t$, which is defined by

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t|\sqrt{1-\beta_t}x_{t-1}, \beta_t \mathrm{I}) \tag{3}$$

where β_t is a small value increasing weakly according to the time step t (typically from 0.0001 to 0.02).

This process continues until reaching x_T, where $T = 1000$ is the total number of steps. At this point, x_T becomes an entirely noisy image. The forward process from x_0 to x_t can be simplified with the Markov chain as

$$\begin{aligned} q(x_t|x_0) &= \prod_{t=1}^{T} q(x_t|x_{t-1}) \\ &= \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t}x_0, \left(1-\bar{\alpha}_t\right)\mathrm{I}\right) \end{aligned} \tag{4}$$

where $\alpha_t = 1-\beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t}\alpha_s$.

The added noise follows a Gaussian distribution, i.e., $\epsilon_{t-1}, \epsilon_{t-2}, \ldots, \sim \mathcal{N}(0,1)$. By using the reparameterization trick, the final forward process can be simplified as

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\cdot\epsilon, \epsilon \sim \mathcal{N}(0,1). \tag{5}$$

2) Backward Denoising Process: The backward process involves training the network and predicting the results. At each iteration, the network takes pairs of produced noisy images and corresponding condition images (x_t, y) as input data. Therefore, the backward process iteratively denoises using the inverse diffusion process

$$p_\theta(x_{t-1}|x_t, y) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, y, t), \sum_\theta(x_t, y, t)). \tag{6}$$

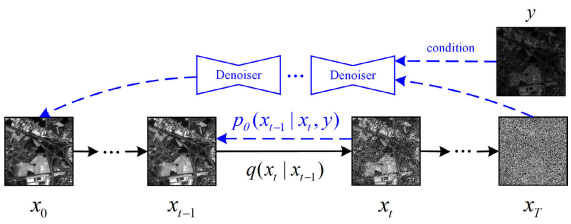


Fig. 1. Forward (black) and backward (blue) diffusion for the SAR image despeckling.

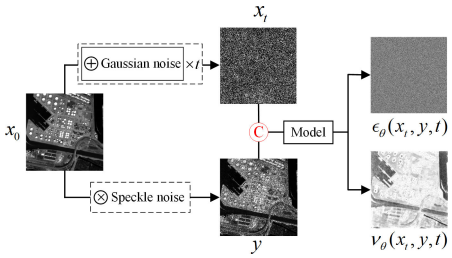


Fig. 2. Framework of forward diffusion for SAR despeckling.

During the training phase, the simulated SAR image y generated by (1) and the noisy image x_t generated by (5) are fed into the network. The network generates two outputs, namely the estimated noise $\epsilon_\theta(x_t, y, t)$ and the variance component $v_\theta(x_t, y, t)$. The workflow is presented in Fig. 2, which converts the despeckling problem into a conditional generation issue such that the diffusion model could be harnessed to restore clean SAR images. During the prediction phase, the real-world SAR image y and the noisy image x_t generated by random Gaussian noise are fed into the model to predict the reference denoised SAR image.

III. Methodology

Some changes are made for the standard diffusion model. A preprocessing is added to convert the multiplicative gamma noise to an approximate Gaussian distribution. An improved the network structure is used. The sampling module is adjusted to accommodate larger reconstructed images. A new loss function is defined incorporating the structural constraints. These improvements will be detailed below.

A. Preprocessing and Postprocessing

A diffusion model learns the distribution of clean data, in which the Gaussian noise is added at each step to partially mask the clean data and guide the network training. For the backward diffusion at step t, the masked information is partially recovered. If a synthetic training dataset is used from optical satellite images, good structures are learned. However, the data distribution between optical and SAR images are not completely consistent as the domain gap exists. Clean SAR images are expected for training when it is possible. In some studies, clean SAR data are approximated by stacking multitemporal data, which is used for fine-tuning neural networks. In this case, necessary preprocessing and postprocessing operations can improve the quality of training.

The logarithmic transformation and the Yeo–Johnson transformation are used for preprocessing. Xiao et al. [19] have

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suggested the logarithmic operation as the preprocessing converting multiplicative noise into additive noise, considering that the logarithmic gamma noise approximates a Gaussian distribution. Then, SAR despeckling problem can be viewed as a Gaussian denoising problem. Xiao et al. [19] views a log-transformed SAR image as a clean image contaminated with a Gaussian noise, which is used as an intermediate image for backward diffusion. In this case, the logarithmic transformation is indispensable for decoupling image and noise. In addition, a trained model can be further trained by stacked multitemporal datasets to approach noise-free images, in which the logarithmic operation converts the remaining speckle noise in stacked data into Gaussian noise. However, the logarithmic gamma noise differs from a normal distribution concerning the tails and kurtosis. In this case, we suggest the Yeo–Johnson transformation for further correction. The Yeo–Johnson transformation [29] can effectively reduce the heteroscedasticity of data and increase its normality by adjusting the shape of the data distribution. Fig. 3 shows the overall transformation for the gamma noise.

Yeo$(\cdot)$ denotes the Yeo–Johnson transformation defined as

$$\text{Yeo}(x) = \begin{cases} \dfrac{(x+1)^{\lambda}-1}{\lambda}, & x \geq 0,\ \lambda \neq 0 \\ \log(x+1), & x \geq 0,\ \lambda = 0 \\ -\dfrac{(-x+1)^{2-\lambda}-1}{2-\lambda}, & x < 0,\ \lambda \neq 2 \\ -\log(-x+1), & x < 0,\ \lambda = 2 \end{cases} \tag{7}$$

where the parameter λ is automatically calculated from the log-transformed gamma noise.

Therefore, to deal with the weak noise in real-world SAR data that may be used for fine-tuning or synthetic training, we propose a combined log-Yeo–Johnson transformation for data preprocessing, which is an reversible transformation. The outcomes of the proposed diffusion model have to experience the inverted Yeo–Johnson transformation and the exponential transformation for postprocessing. The inverse of Yeo–Johnson transformation is defined as

$$\text{Yeo}^{-1}(x) = \begin{cases} (\lambda x+1)^{1/\lambda}-1, & x \geq 0, \lambda \neq 0 \\ e^{x}-1, & x \geq 0, \lambda = 0 \\ -[(\lambda-2)x+1]^{1/(2-\lambda)}, & x < 0, \lambda \neq 2 \\ -e^{-x}+1, & x < 0, \lambda = 2 \end{cases} \tag{8}$$

where λ is consistent with those in preprocessing.

In order to adapt to the multiplicative scattering noise characteristics in stacked multitemporal SAR images, the diffusion model proposed in this article will not use the additive Gaussian noise, but adopt the multiplicative gamma noise processed by the log-Yeo–Johnson transformation for diffusion. The transformation makes the model suitable for a second training with real-world SAR images. To this end, the proposed method performs log-Yeo–Johnson transformations on synthetic datasets built with clean optical images, too. The reference image y is not involved in the transformation. This preprocessing ensures the consistency of the processed noise between the initial training and fine-tuning. When the

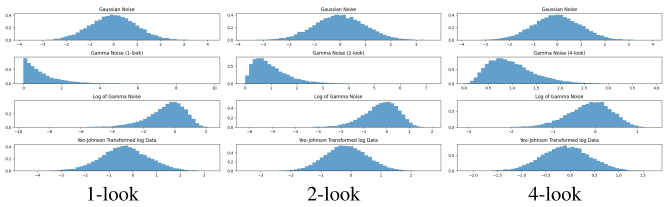


Fig. 3. Log-Yeo–Johnson transformation for simulated gamma noise with various equivalent numbers of looks.

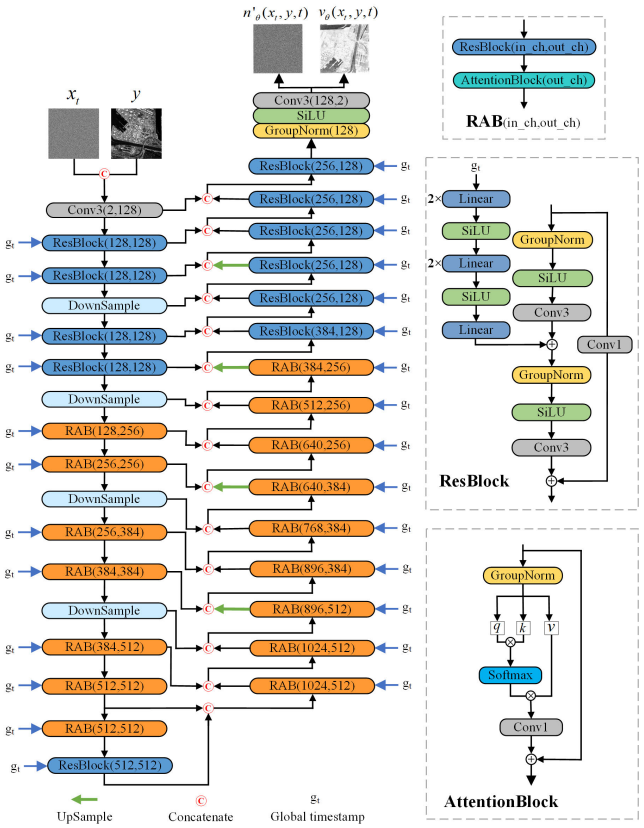


Fig. 4. Network structure for noise estimation. The numbers of input and output channels are marked for each block. Conv3 denotes the convolution with the kernel size of 3 × 3. Conv1 denotes the convolution with 1 × 1 kernels. ⊕ denotes the add operation. ⊗ denotes the multiplication operation.

transformation is used, (5) is modified to

$$x_t = \sqrt{\bar{\alpha}_t}x_0' + \sqrt{1-\bar{\alpha}_t}\cdot n' \tag{9}$$

where

$$x_0' = \text{Yeo}(\log(x_0)) \tag{10}$$

$$n' = \text{Yeo}(\log(n)). \tag{11}$$

B. Proposed Network Structure

The network structure of the proposed diffusion model is shown in Fig. 4. Referenced by Ho et al. [26], the codec framework is used for denoising which consists of an encoding part, a decoding part, and skip connections similar to U-Net.

As shown in the left branch of Fig. 4, the encoding part consists of one 3 × 3 convolutional (conv3) layer, five residual blocks (ResBlocks), seven residual and attention blocks (RABs), and four downsampling. The decoding part is

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composed of nine RABs, six ResBlock, and four upsampling blocks (UpBlocks), a group normalization (GroupNorm) layer, a sigmoid linear unit (SiLU) activation layer, and a 3×3 convolutional layer. A noisy image x_t and an observed SAR image y are fed into the network, which outputs the estimated less noisy image and the variance of the noise. Skip connections are used between the encoding and decoding parts preserving details due to the downsampling operation. The downsampling block is a 3×3 convolution with a stride of 2. The UpBlock in the decoding part contains a nearest neighbor interpolation and a 3×3 convolutional layer. The numbers of input and output channels are given in Fig. 4.

A ResBlock in Fig. 4 consists of two convolutional layers for feature extraction, whereas GroupNorm and SiLU are also used. In addition, the global time stamp g_t is processed through a series of linear and activation layers before being injected into the ResBlock. The global time stamp g_t records the time stamps ready for the backward diffusion. The conversion from t to g_t can be referenced from [30].

Besides the ResBlock, an attention block (AttentionBlock) in Fig. 4 is also involved in each RAB block. AttentionBlock uses the self-attention mechanism. The input vector is normalized with GroupNorm to output the query (q), key (k), and value (v) vectors. Then, the attention weight is calculated through matrix multiplication and a softmax function, which gives the weight for the value branch. A 1×1 convolution (Conv1) is performed on the weighted v, which is added on the input to give the output of the AttentionBlock.

The variance learning strategy in [30] is used to improve the model training. Based on the obtained $n'_\theta(x_t, y, t)$ and $v_\theta(x_t, y, t)$ from the model prediction, the mean value $\mu_\theta(x_t, y, t)$ and variance $\sum_\theta(x_t, y, t)$ in (6) are calculated by

$$\mu_\theta(x_t, y, t) = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} n'_\theta(x_t, y, t)\right) \tag{12}$$

$$\sum_\theta(x_t, y, t) = \frac{v_\theta(x_t, y, t)+1}{2}\log\beta_t + \left(1 - \frac{v_\theta(x_t, y, t)+1}{2}\right)\log\widetilde{\beta}_t \tag{13}$$

where the noise $n'_\theta(x_t, y, t)$ and the variance component $v_\theta(x_t, y, t)$ are learned and predicted by the modified model.

C. Sampling for Backward Diffusion

To accelerate the sampling operation, the deterministic sampling strategy in the denoising diffusion implicit model [32] is adopted, which sets $\sum_\theta(x_t, y, t) = 0$ in the test stage. The output of the network gives the estimated noise image $n'_\theta(x_t, y, t)$, which is used to predict the recovered image iteratively through the backward diffusion as

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\left(\frac{x_t - \sqrt{1-\bar{\alpha}_t}\cdot n'_\theta(x_t, y, t)}{\sqrt{\bar{\alpha}_t}}\right) + \sqrt{1-\bar{\alpha}_{t-1}}\cdot n'_\theta(x_t, y, t). \tag{14}$$

During the testing phase, the real-world SAR image y and the log-Yeo–Johnson transformed gamma noise image n' are fed into the network to obtain the denoised image x. To address arbitrary sizes and potential artifacts, a patch-based sampling technique is proposed. Usually, a large image is divided into smaller patches for restoration, and then the results are combined for mosaicking. This strategy, however, may suffer from edge artifacts because a pure noise patch is needed as the input of the network in each iteration. In cases where there are different levels of noise for the same image content, the resulting mosaic may show shadowed edges. To eliminate the artifacts, the pure noise image is generated and cropped accompanied with the image patches. Fig. 5 demonstrates the process of patch-based testing, where the noise patches are extracted simultaneously with the image patches.

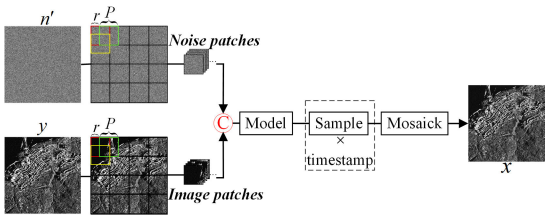


Fig. 5. Overlapped-patch-based test for arbitrary image sizes.

D. Loss Functions

To train the proposed diffusion model, a combination of the mean square error (MSE) and the Kullback–Leibler (KL) divergence is used as the loss function to evaluate the noise fitness

$$\mathcal{L}_{\text{train}} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{KL}}\cdot\mathcal{L}_{\text{KL}} \tag{15}$$

$$\mathcal{L}_{\text{MSE}} = \frac{1}{M}||n' - n'_\theta(x_t, y, t)||_2^2 \tag{16}$$

$$\mathcal{L}_{\text{KL}} = D_{\text{KL}}\left(q\left(x_{t-1}|x_t, x'_0\right)||p_\theta(x_{t-1}|x_t, y)\right) \tag{17}$$

where M denotes the quantity of training image pairs, $D_{\text{KL}}(\cdot)$ is used to calculate the KL divergence between two distributions, and λ_{KL} is an adjustable parameter balancing the losses. λ_{KL} is set to 1×10^4 in the experiment. $p_\theta(x_{t-1}|x_t, y)$ is calculated by (6). $q(x_{t-1}|x_t, x'_0)$ is the posterior conditional probability, which is

$$q\left(x_{t-1}|x_t, x'_0\right) = \mathcal{N}\left(x_{t-1}; \tilde{\mu}_t\left(x_t, x'_0\right), \tilde{\beta}_t \mathrm{I}\right) \tag{18}$$

where mean value $\tilde{\mu}_t(x_t, x'_0) = ((\bar{\alpha}_{t-1})^{1/2}\beta_t/(1-\bar{\alpha}_t))x_0{}' + (\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1}))/(1-\bar{\alpha}_t)x_t$, and variance $\tilde{\beta}_t = (1-\bar{\alpha}_{t-1})/(1-\bar{\alpha}_t)\beta_t$.

The loss function for validation differs from that for training because the training loss evaluates the produced noise whereas the validation evaluates the recovered images. In validation, the MSE $\mathcal{L}_{\text{MSE}}$, the despeckling gain (DG) loss $\mathcal{L}_{\text{DG}}$ [31], and the TV loss $\mathcal{L}_{\text{TV}}$ are integrated. The loss function for validation is defined as

$$\mathcal{L}_{\text{validation}} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{DG}}\cdot\mathcal{L}_{\text{DG}} + \lambda_{\text{TV}}\cdot\mathcal{L}_{\text{TV}} \tag{19}$$

$$\mathcal{L}_{\text{MSE}} = \|x - x_0\|_2^2 \tag{20}$$

$$\mathcal{L}_{\text{DG}} = \log_{10}\frac{\|x - x_0\|_2^2}{\|y - x_0\|_2^2} \tag{21}$$

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$$\mathcal{L}_{\mathrm{TV}} = \sum_{i=1}^{W}\sum_{j=1}^{H}\sqrt{\left(x_{i+1,j}-x_{i,j}\right)^2+\left(x_{i,j+1}-x_{i,j}\right)^2} \quad (22)$$

where x_0 is a clean image without the log-Yeo–Johnson transformation, y is the speckled (SAR) image, x is the despeckling result, W and H are the width and height of the images, respectively, λ_{DG} and λ_{TV} are adjustable parameters. In the experiment, the values of λ_{DG} and λ_{TV} are 1 and 10, respectively.

IV. Experimental Scheme

In the experimental procedure, the newly proposed DiffusionSAR method is trained on two datasets and tested using four sets of images. The results are compared with six state-of-the-art methods. The datasets and parameters for training and testing are provided in this section, along with the metrics used to evaluate the effectiveness of denoising.

A. Datasets for Training and Testing

Due to the unavailability of clean SAR images, it is widely accepted that SAR images are synthesized from optical images for training deep-learning-based methods. These models learn the despeckling process in simulated images and then apply it to real-world SAR images. Following this strategy, two optical datasets were used for training, including the UC Merced (UCM) dataset [32] and a newly designed dataset.

The UCM dataset is used for land use recognition. It has 21 categories of covering farmland, airplane, baseball diamond, beach, buildings, chaparral, dense residential, forest, freeway, golf course, harbor, intersection, medium residential, mobile home park, overpass, parking lot, river, runway, sparse residential, storage tanks, and tennis court. Each category consists of 100 airborne images in the size of 256 × 256. The ground resolution is approximately 0.3 m.

The resolution values of satellite SAR images are much higher than 0.3 m, so the models trained by the UCM dataset may not adapt well. Therefore, a new training dataset is specifically constructed for satellite SAR despeckling in order to reduce the scale difference. The dataset consists of 2400 nonoverlapping images with dimensions of 256 × 256 pixels and a ground resolution of 10 m. These images were captured by Sentinel-2 and cropped across 16 locations. Four main categories are grouped, including City, Farm, Forest, and Port, with 600 images in each category. This dataset is named Sentinel-2's Four Scenes (S2FS). The data sources for the newly constructed dataset are listed in Table I.

The datasets are partitioned for training, validation, and testing. For each scene of the UCM dataset, 80 images are for training, 15 images are for validation, and five images are for testing. For each scene in the newly proposed dataset, 500 images are for training, 90 images are for validation, and ten images are for testing. To simulate SAR images with diverse noise levels, the clean images were multiplied by speckle, where the numbers of equivalent looks were set to one, two, and four, respectively.

Real-world data are also used for test. The real-world SAR images are from the ALOS, Sentinel-1, and TerraSAR-X satellites. Each source contains a single-look intensity SAR image of a City scene and a Farm scene with a unified size of 512 × 512 pixels.

TABLE I
Data Sources and Main Features of the New Dataset

location (city)	captured time	scenes
Smederevo	20 Jun 2022	Farm, Forest
Jiamusi	17 May 2023	Forest
Oradea	20 Jun 2022	Farm
Timisoara	30 Jun 2022	Farm, Forest
Brasov	20 May 2022	Forest
Annapolis	25 May 2023	Port
Hong Kong	28 Feb 2023	Port
Shanghai	28 Apr 2023	Port
Zhoushan	10 Apr 2023	Port
Suzhou	14 Mar 2023	City, Farm
Shaoxing	14 Mar 2023	City, Port
Rotterdam	13 Aug 2022	Port
Leuven	1 Jun 2023	City
Paris	13 Aug 2022	City
Rouen	3 May 2023	City, Farm
Limoges	9 Jul 2022	City, Farm

B. Methods for Comparison

The new method is compared with eight SAR despeckling algorithms, including PPB [6], FANS [8], SAR2SAR [22], Speckle2Void [23], MONet [13], SAR-CAM [16], TransSAR [17], and SAR-ON [33]. Among them, PPB and FANS are nonlocal methods, and the remaining methods are neural-network-based methods. Specifically, SAR2SAR and Speckle2Void are self-supervised algorithms free of training. MONet and SAR-ON are based on CNNs, SAR-CAM uses continuous AttentionBlocks, and TransSAR uses transformer. MONet, SAR-CAM, TransSAR, and SAR-ON were all trained using the same dataset as the method proposed.

Parameters are set for competing algorithms. Nonlocal methods need to specify the window and search range. For PPB, the patch size is 7 × 7 and the search range is 21 × 21. For FANS, the patch size is 8 × 8 and the search range is 39 × 39. Neural-network-based methods kept their default parameter settings. SAR2SAR and Speckle2Void are self-supervised algorithms that are free of online training. The parameters of SAR2SAR was trained with the Sentinel-1 dataset in [22]. The parameters of Speckle2Void was trained with the TerraSAR-X dataset in [23]. A decorrelation preprocessing is needed in Speckle2Void, in which the parameters were carefully set image by image for good performance. The nonlocal methods are programmed in MATLAB and the others are in Python.

The parameters of our method are also given. The total number of epochs is set to 600. The size of the input is 128 × 128. The batch size is 20 with the learning rate $2e-5$. Adam is used as the optimizer with parameter $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 1e-8$. The code was programmed in Python, where the PyTorch framework was used for deep learning. Training and test were carried out on NVIDIA Tesla A100 for deep learning of neural-network-based methods.

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C. Metrics

Metrics are used to evaluate the despeckling results. For the results on the simulated data, the peak signal-to-noise ratio (PSNR) and the structural similarity index (SSIM) are used to evaluate the distance between the results and the clean images. For the results on the real-world data where the speckle-free SAR images are missing, nonreference metrics are used, including the edge preservation degree based on the ratio of the average (EPD-ROA) [34], the β-ratio index of ratio image (β_{ratio}) [35], the statistical quality index (SQI) [36], the preservation of mean values (Mean), the equivalent number of looks (ENL) [37], the mean of image (MOI) [38], and the root MSE of strong scatters (RMSE-SS).

Metrics are introduced here. EPD-ROA measures the ability of edge consistency and can be divided into the horizontal and vertical scores. The closer the value of EPD-ROA approaches 1, the richer the edges are preserved. The β_{ratio} index calculates the normalized correlated coefficient between the ratio image and the original SAR image being high-pass filtered. The ratio image is the ratio between the original SAR image and the despeckled image (y/x). β_{ratio} is expected to be as small as possible because a ratio image is considered as noncorrelated to the original SAR image. The SQI denotes the statistical similarity of the ratio between the noisy and despeckled images. SQI ranges from 0 to 1 and a higher value indicating better histogram matching. The ENL index describes the level of speckle. A higher ENL value indicates a lower speckle level in the SAR image and the better performance of speckle reduction. Giving a homogeneous area in the original and despeckled images, MOI denotes the ratio of the mean values. The closer the MOI is to 1, the better the filter is considered. The RMSE-SS is tested in the real-world experiment by choosing strong scatters and calculating the RMSE change before and after despeckling. The mean values (Mean) before and after despeckling are expected to be consistent, which are presented in the real-world comparison to illustrate the preservation of mean values.

V. Experimental Results of Synthetic Scenes

The proposed method and comparison methods were tested on four types of synthetic scenes involving 40 images, and the results are presented in this section. The newly designed S2FS dataset is used for this test. For each scene, average scores of ten test images are presented in Table II to observe the structural and radiometric changes. The optimal values are highlighted in bold. For each scene, three images are manifested with regard to the simulated 1-, 2-, and 4-look noise levels as can be found in Figs. 6–8, respectively. SAR2SAR and Speckle2Void are not tested on 2- and 4-look simulation because they are unsupervised methods trained on 1-look scenes.

Fig. 6 shows the despeckling results of a simulated 1-look City scene, where the simulated noisy image is severely corrupted by speckles. The results of nonlocal methods (PPB and FAN) are smooth for buildings, but artifacts are observed and details are missing. MONet and SAR-CAM fail to effectively remove speckle. Less noise is observed for TransSAR and SAR-ON, but the structures of buildings are distorted. In contrast, the proposed method maintains clear shapes for buildings, minimizes deformation in homogeneous areas, and recovers the richest details compared to the reference image.

TABLE II
Evaluation on the Simulated Data From Sentinel-2

1-look	Farm		City		Forest		Port	
	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
PPB	0.8054	28.96	0.6109	19.83	0.8652	34.75	0.6934	21.81
FANS	0.8426	29.58	0.6994	21.29	0.8941	36.05	0.7566	23.13
SAR2SAR	0.8032	27.96	0.6442	20.19	0.8166	32.61	0.6791	20.57
Speckle2Void	0.8049	29.01	0.6417	19.40	0.8296	33.86	0.7266	20.70
MONet	0.8007	28.65	0.6971	20.92	0.8033	33.09	0.7090	22.58
SAR-CAM	0.8004	28.90	0.7188	21.54	0.8949	36.55	0.7711	23.35
TransSAR	0.8079	29.03	0.7110	21.46	0.8780	35.73	0.7594	23.27
SAR-ON	0.8245	29.38	0.7078	21.56	0.8386	35.14	0.7578	23.23
proposed	**0.8701**	**30.05**	**0.7382**	**22.33**	**0.9050**	**36.57**	**0.7928**	**23.38**
2-look	Farm		City		Forest		Port	
	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
PPB	0.8496	30.54	0.7306	21.45	0.8942	36.06	0.7833	23.37
FANS	0.8769	31.57	0.7960	23.11	0.9216	37.63	0.8352	24.94
MONet	0.8229	30.42	0.8042	23.18	0.9205	37.73	0.8336	25.03
SAR-CAM	0.8356	30.41	0.7987	22.78	0.9279	38.10	0.8359	24.61
TransSAR	0.8294	30.21	0.7897	23.07	0.8934	36.69	0.8239	24.80
SAR-ON	0.8522	30.73	0.7812	23.08	0.8926	36.71	0.8214	24.87
proposed	**0.8915**	**31.71**	**0.8113**	**23.28**	**0.9285**	**38.53**	**0.8493**	**25.14**
4-look	Farm		City		Forest		Port	
	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
PPB	0.8812	31.87	0.8101	23.03	0.9155	37.28	0.8481	24.92
FANS	0.9060	33.38	0.8649	25.06	0.9430	39.24	0.8897	26.79
MONet	0.8899	32.88	0.8750	25.55	0.9244	38.84	0.8930	26.22
SAR-CAM	0.8645	31.94	0.8599	25.08	0.9488	39.81	0.8861	26.75
TransSAR	0.8806	32.05	0.8393	24.74	0.8881	36.93	0.8588	26.26
SAR-ON	0.8831	32.23	0.8435	24.83	0.8861	36.89	0.8668	26.41
proposed	**0.9127**	**33.50**	**0.8770**	**25.57**	**0.9562**	**40.38**	**0.9054**	**27.19**

Fig. 7 presents the challenge of the Farm scene from a simulated 2-look contamination. Speckle is removed by techniques such as PPB, FANS, and Speckle2Void, but these methods may result in a loss of details. Artifacts are produced by SAR-CAM, TransSAR, and SAR-ON where the noise are not removed successfully. Only our method can remove simulated noise and maintain clear edges well.

Fig. 8 displays the despeckling results of the Port scene from a synthetic 4-look deterioration. The results of SAR2SAR and Speckle2Void are oversmoothed. The results of PPB and FANS are clear but artifacts are observed. MONet, SAR-CAM, TransSAR, and SAR-ON fail to remove speckle effectively. Our method gives the cleanest result and maintains details reasonably.

Table II presents the evaluation results of the simulated 1-, 2-, and 4-look SAR scenes from the S2FS dataset. Each digit is the average value of a scene by averaging the scores of ten despeckled images. The PSNR and SSIM scores indicate that the newly proposed method outperforms all other methods for all scenes. The structural similarity of our method is significantly higher than competing methods. SAR2SAR, Speckle2Void, and PPB fail to present competitive scores. FANS works well for the homogeneous scenes (Farm and Forest) but poorly for heterogeneous scenes (City and Port). Among the deep-learning-based methods, TransSAR and SAR-ON obtain more admirable results than MONet and SAR-CAM, but their structural scores are far from satisfactory.

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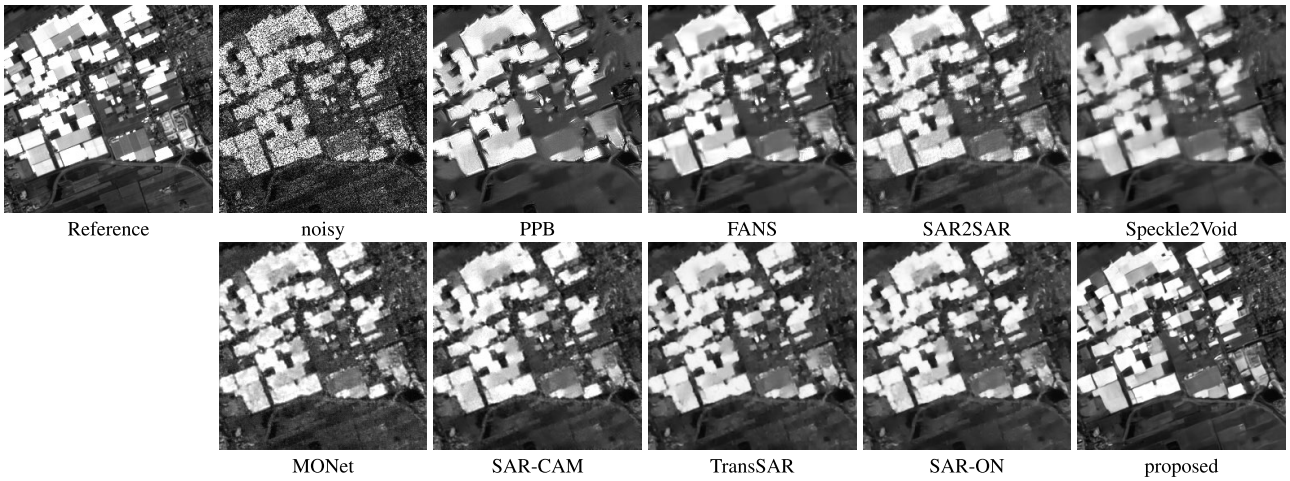


Fig. 6. Despeckling results tested and trained with the simulated 1-look data.

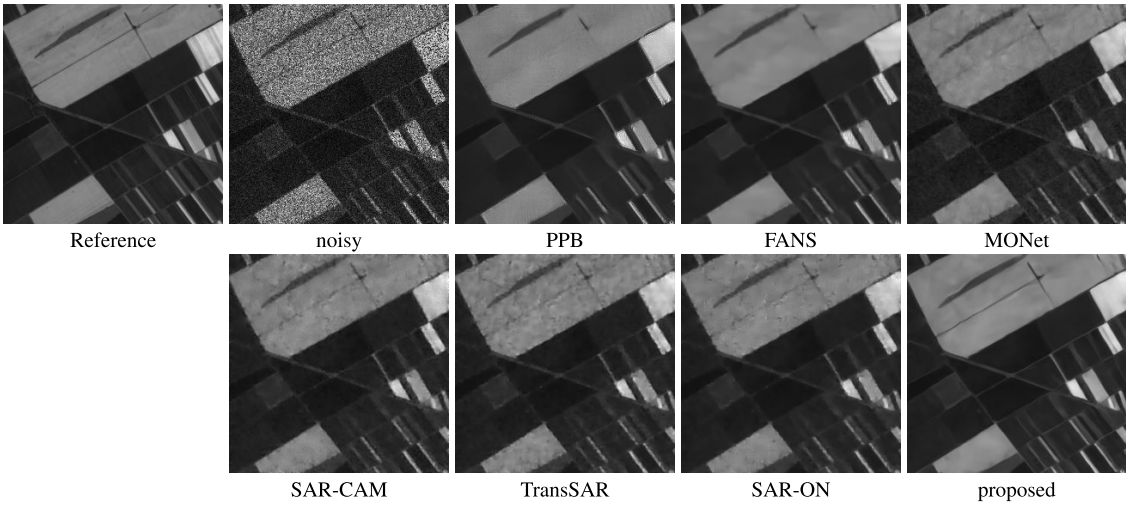


Fig. 7. Despeckling results tested and trained with the simulated 2-look data.

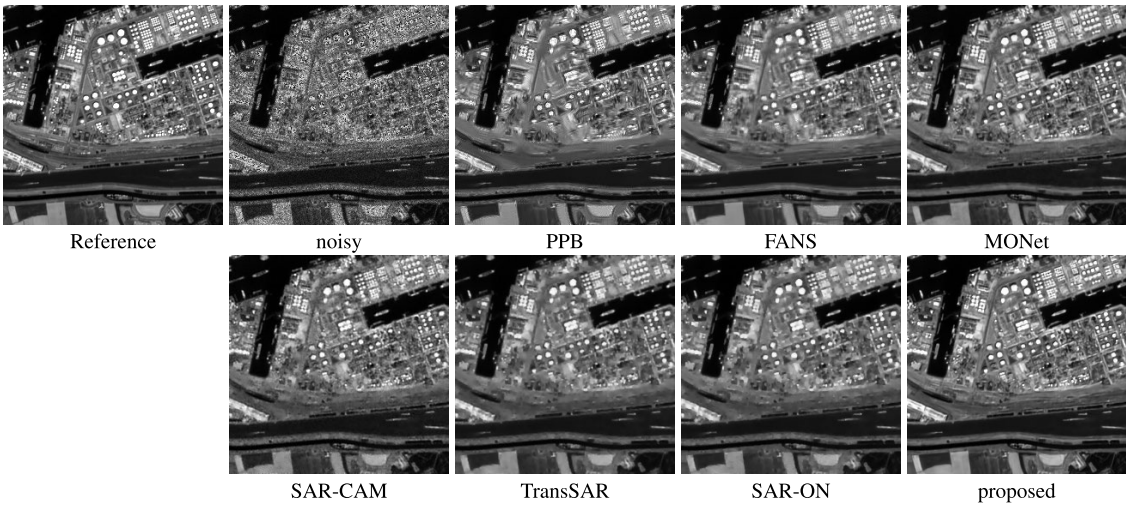


Fig. 8. Despeckling results tested and trained with the simulated 4-look data.

VI. Experimental Results of Real-World Scenes

The proposed method and comparison methods were tested on real-world SAR scenes from three satellite sources, and the results are presented in this section. The measured scores of six real-world single-look complex (SLC) SAR images are listed in tables. To save space, only the results of the City scene are manifested in figures, along with their ratio images.

The despeckling results and their ratio images are presented in Figs. 9–14. The first images of Figs. 9, 11, and 13 display the City scenes that need to be restored, which are taken from ALOS, Sentinel-1, and TerraSAR-X, respectively. The

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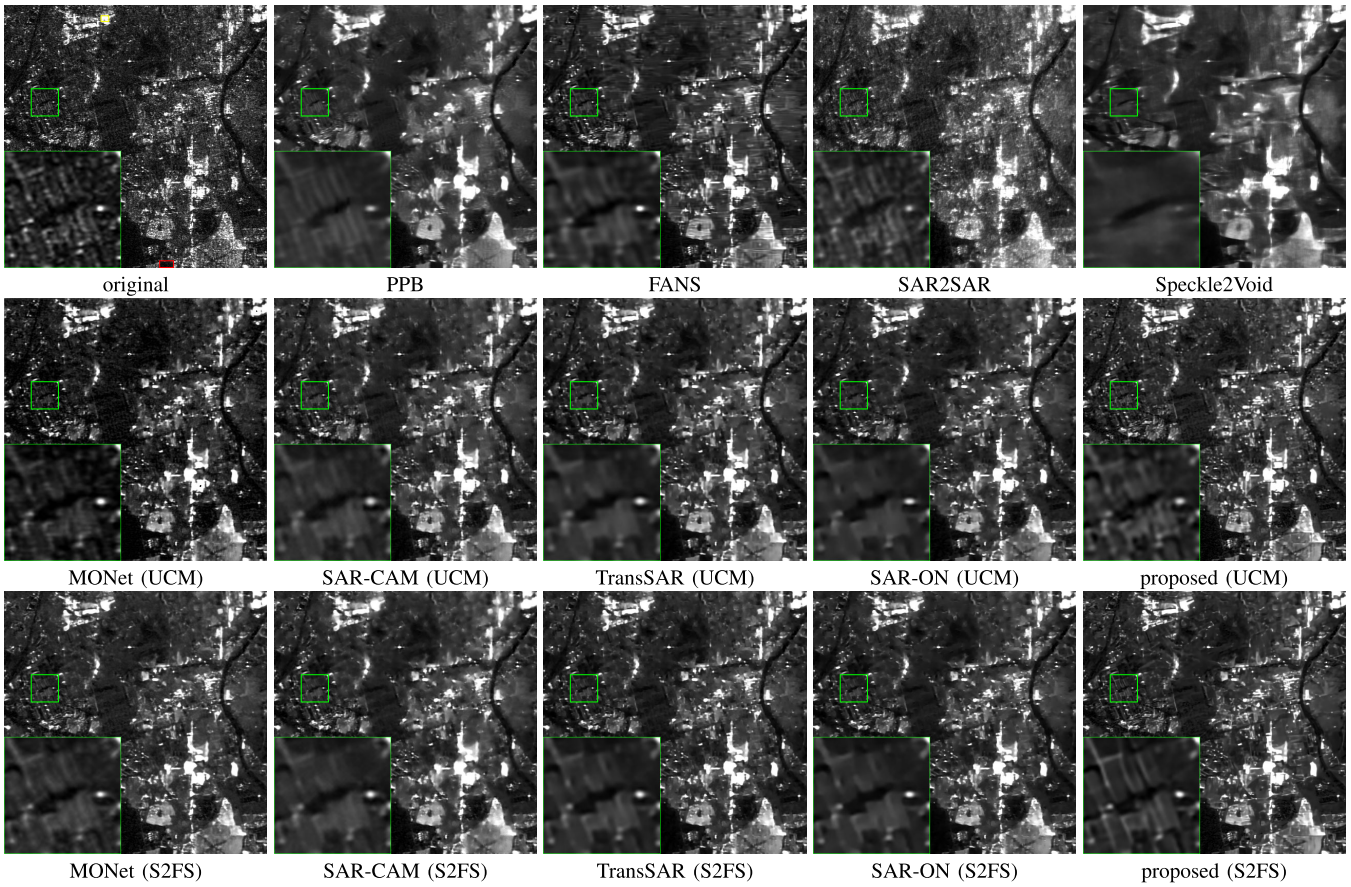


Fig. 9. Despeckling results of the ALOS's City scene.

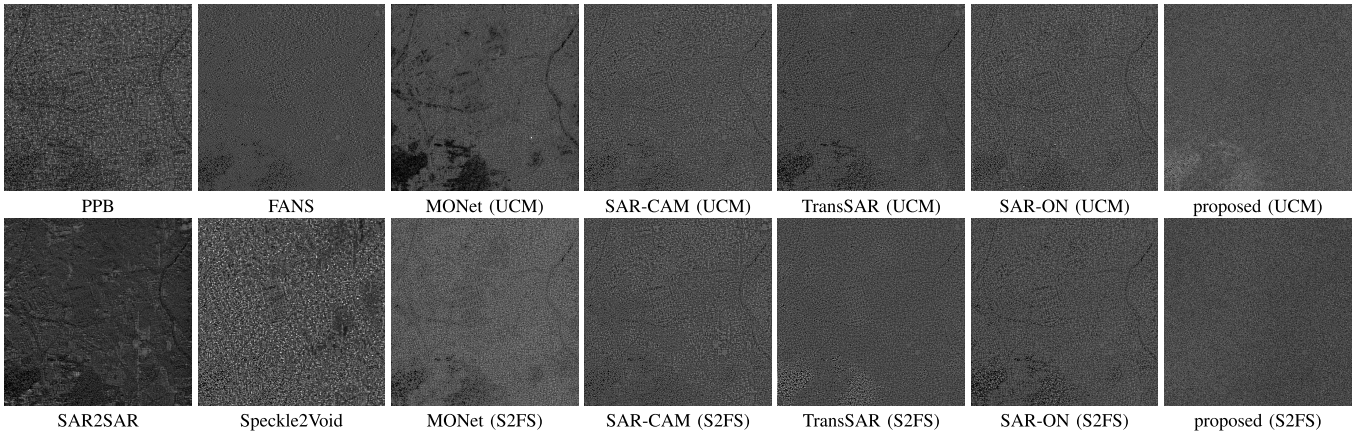


Fig. 10. Despeckling ratio images of the ALOS's City scene.

red boxes in these images indicate the selected homogeneous blocks evaluated by the ENL and MOI metrics. Their sizes are 15×25 in the ALOS image, 25×20 in the Sentinel-1 image, and 15×25 in the TerraSAR-X image. The yellow boxes indicate the areas with strong scatters for the RMSE-SS evaluation. The yellow boxes occupy 9×13 pixels in the ALOS image, 16×16 pixels in the Sentinel-1 image, and 10×13 pixels in the TerraSAR-X image.

Fig. 15 displays the Farm scenes to be restored, taken from the SAR sensors in ALOS, Sentinel-1, and TerraSAR-X, respectively. These scenes contain identifiable homogeneous areas. The red boxes in these images indicate the selected homogeneous areas, which will be evaluated using the ENL and MOI metrics. The selected homogeneous areas have 30×30 pixels in the ALOS image, while it has 35×35 pixels in the Sentinel-1 image and 25×35 pixels in the TerraSAR-X image. The yellow boxes mark the areas of strong scatters for the RMSE-SS evaluation. The sizes are 12×12 for ALOS, 12×27 for Sentinel-1, and 11×7 for TerraSAR-X.

A. Visual Comparison of Real-World City Scenes

The real-world SAR images dedicated to the City scene were despeckled using the proposed and state-of-the-art competing methods. The restored images are shown in Figs. 9–13. The ratio images are shown in Figs. 10–14.

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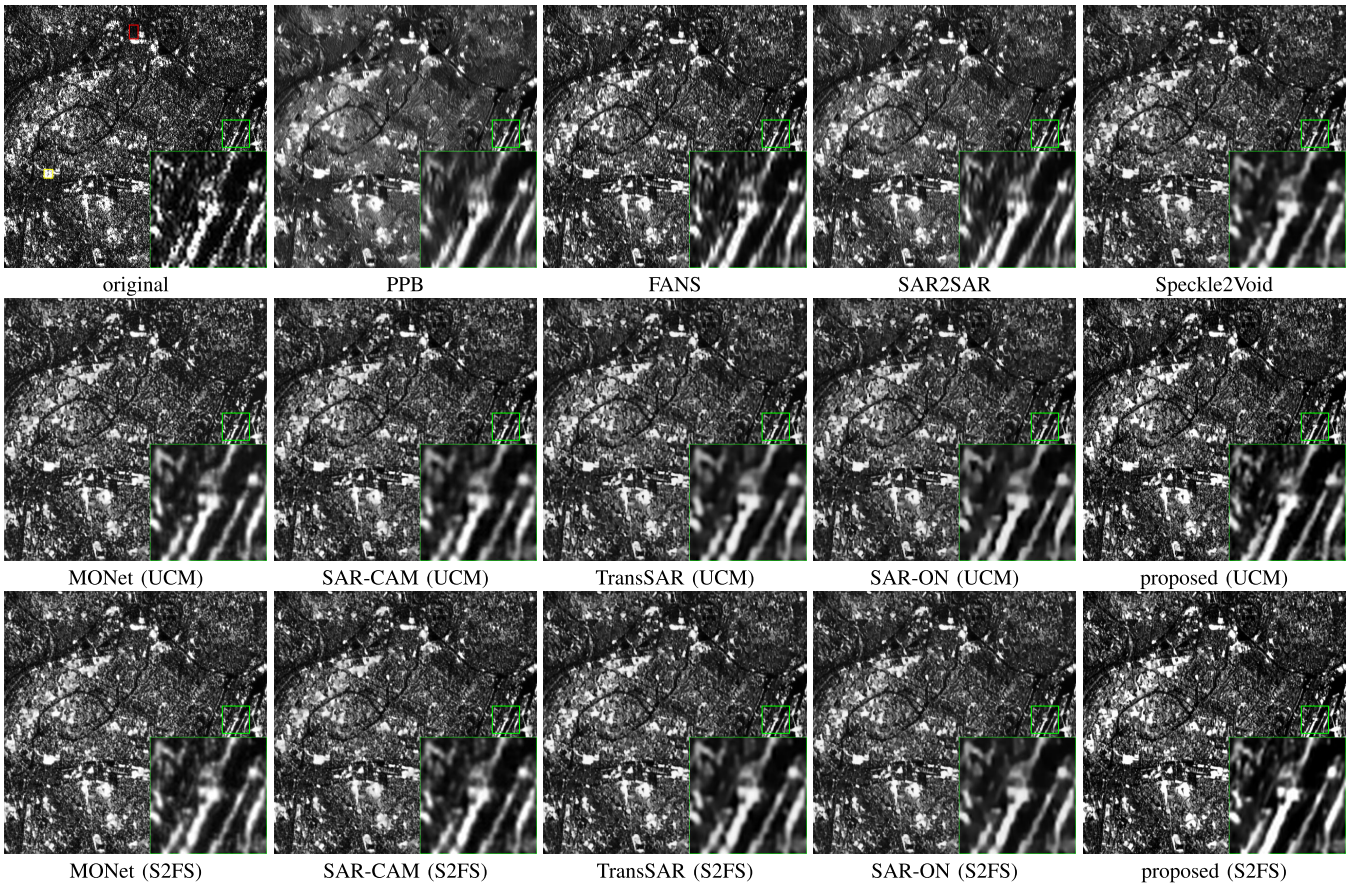


Fig. 11. Despeckling results of the Sentinel-1's City 1 scene.

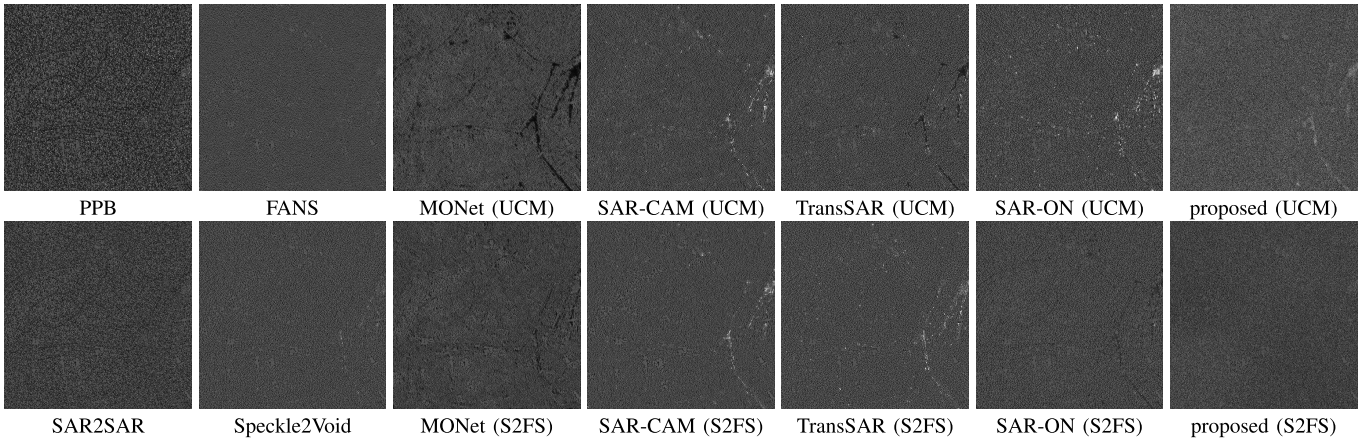


Fig. 12. Despeckling ratio images of the Sentinel-1's City 1 scene.

Fig. 9 presents the despeckling results of SAR images acquired by ALOS PALSAR in HH polarization, with a spatial resolution of 10 × 10 m, specifically focusing on the City scene. The results obtained from PPB and FANS exhibit excessive smoothing, resulting in noticeable artifacts. The result by Speckle2Void is oversmoothed. The results of SAR2SAR and MONet fail to remove speckles effectively. Generally, results from models trained by S2FS yield richer structures than those trained by UCM. The proposed model trained by S2FS retains more visually acceptable details.

The ratio images of the ALOS's City scene are shown in Fig. 10. An ideal filter should generate a ratio image that consists solely of speckle. Therefore, the effectiveness of a filter can be judged by the amount of correlations and structures preserved in the ratio image. Among all the evaluated methods, FANS demonstrates the least structures in the ratio image, whereas TransSAR and our method (trained by S2FS) work well, too.

Fig. 11 presents the despeckling results of SAR images acquired by Sentinel-1 in VV polarization, with a spatial resolution of 3.5 × 22 m, specifically focusing on the City scene. Both artifacts and noise are noticeable in the results obtained from PPB, FANS, and SAR2SAR. Speckle2Void produces oversmoothness again. Supervised

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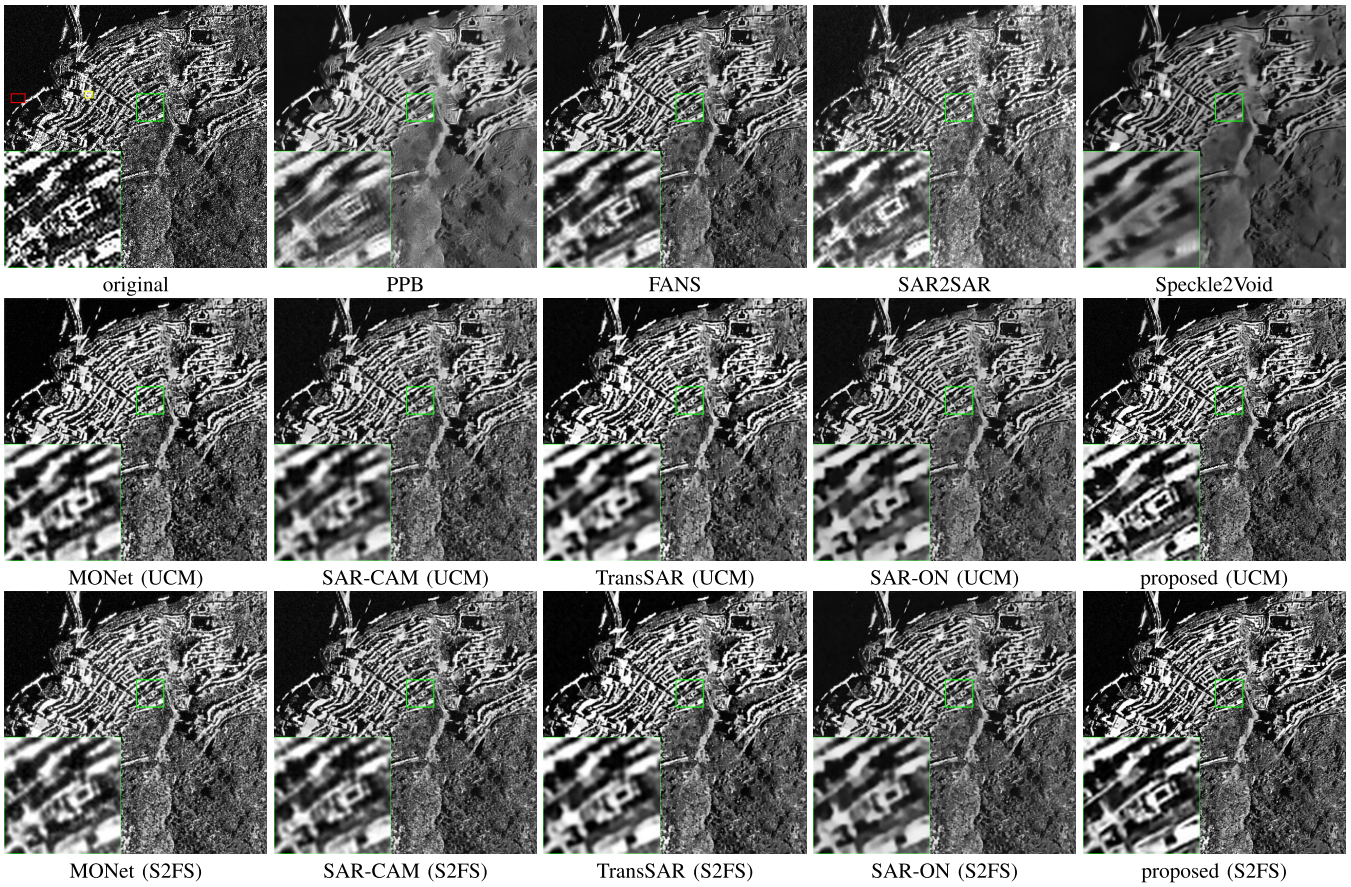


Fig. 13. Despeckling results of the TerraSAR-X's City scene.

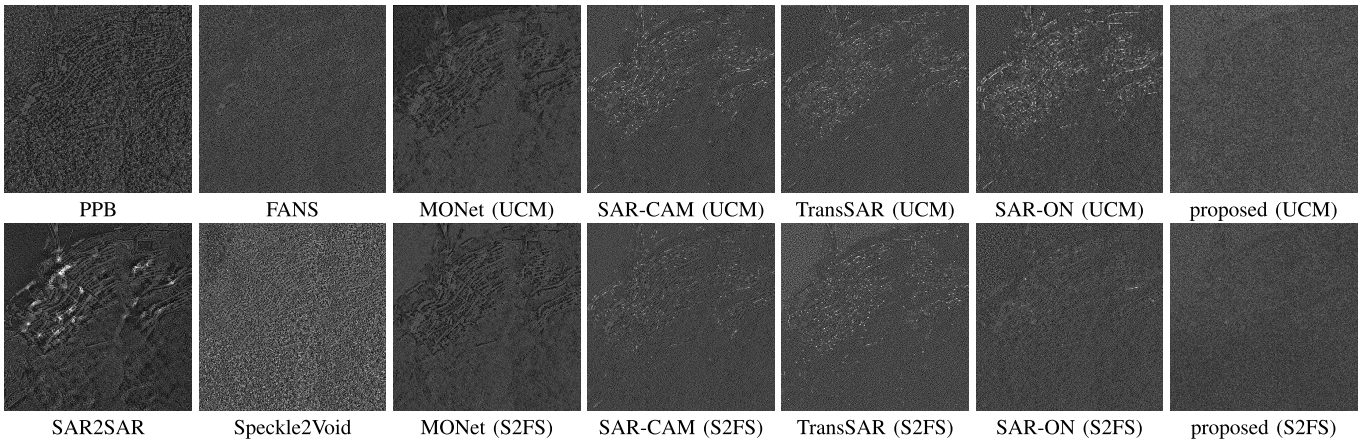


Fig. 14. Despeckling ratio images of the TerraSAR-X's City scene.

deep-learning-based approaches yield comparable results. The proposed method preserves the richest details. It is worth noting that the influence of training sources is minimal because the structures exhibited in the despeckled images from various training datasets are remarkably similar.

The ratio images obtained during the despeckling process of the Sentinel-1's City image are shown in Fig. 12. Among all the methods, both FANS and the proposed method trained by S2FS exhibit the fewest structures in the ratio images. However, the ratio images produced by the proposed method trained by UCM demonstrate clear structural patterns.

Fig. 13 presents the despeckling results of SAR images acquired by TerraSAR-X in HH polarization, with a spatial resolution of 4.02 $\times$ 4.03 m, specifically focusing on the City scene. In the results produced by PPB, FANS, and Speckle2Void, ringing artifacts and oversmoothness are observed. The despeckling results from MONet and TransSAR exhibit noticeable speckles. On the other hand, the results obtained from SAR-CAM, SAR-ON, and our proposed method display similar performance. Notably, the models trained by S2FS retain more detailed information than those trained by UCM.

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TABLE III
EVALUATION OF THE REAL-WORLD CITY SCENE

	train set	EPD-ROA hori	EPD-ROA vert	$\beta_{\text{ratio}}\downarrow$	SQI↑	ENL↑	MOI $\rightarrow 1$	RMSE -SS	Mean
ALOS PALSAR (mean value: 43.11)									
PPB	-	0.9588	0.9855	0.0832	0.7952	124.27	0.8347	36.39	47.95
FANS	-	0.9662	**0.9992**	0.0164	0.8485	81.39	0.8927	35.71	48.76
SAR2SAR	-	0.9602	0.9893	0.1223	0.8052	104.81	0.9048	35.78	**44.33**
Speckle2Void	-	0.9543	0.9834	0.0318	0.8291	82.08	0.9005	38.75	45.92
MONet	UCM	0.9660	**0.9992**	0.0790	0.8002	120.82	0.7971	36.18	47.42
	S2FS	0.9696	0.9927	0.0309	0.8454	100.69	0.9096	34.34	46.15
SAR-CAM	UCM	0.9655	0.9935	0.0909	0.8635	105.00	0.9073	40.18	45.60
	S2FS	0.9641	0.9932	0.0592	0.8631	103.64	0.9127	44.27	45.69
TransSAR	UCM	0.9605	0.9906	0.0389	0.8199	106.26	0.8438	34.35	49.50
	S2FS	0.9634	0.9924	0.0530	0.8570	97.35	0.8883	33.46	47.97
SAR-ON	UCM	0.9620	0.9898	0.0548	0.8500	106.39	0.9021	35.57	45.68
	S2FS	0.9612	0.9908	0.0406	0.8465	111.08	0.8835	34.84	46.49
Ours	UCM	0.9716	0.9902	0.0279	**0.8851**	102.34	0.9384	35.38	45.46
	S2FS	**0.9765**	0.9936	**0.0062**	0.8542	**129.65**	**0.9428**	**32.73**	47.59
Sentinel-1 (mean value: 64.53)									
PPB	-	0.9456	0.8589	0.2713	0.6386	76.48	0.7863	57.40	79.89
FANS	-	1.0459	0.8925	0.0185	0.8163	59.87	0.8864	56.07	72.59
SAR2SAR	-	0.9481	0.8680	0.1961	0.7056	69.67	0.8279	56.53	76.49
Speckle2Void	-	0.9493	0.8790	0.0603	0.7502	65.33	0.8919	55.36	72.33
MONet	UCM	0.9628	0.8850	0.2100	0.7296	60.35	0.7843	55.65	75.33
	S2FS	0.9765	0.8944	0.1640	0.7378	71.99	0.7829	56.77	77.08
SAR-CAM	UCM	0.9447	0.8792	0.1255	0.7421	61.94	0.8886	56.62	73.89
	S2FS	0.9580	0.8885	0.0978	0.7624	64.65	0.8856	55.28	72.73
TransSAR	UCM	0.9413	0.8729	0.1219	0.7308	61.84	0.8659	55.48	73.90
	S2FS	0.9477	0.8794	0.0994	0.7562	70.78	0.9134	55.31	72.45
SAR-ON	UCM	0.9366	0.8742	0.1245	0.7322	68.62	0.8543	55.59	71.82
	S2FS	0.9454	0.8755	0.0926	0.7343	68.95	0.8777	55.75	71.81
Ours	UCM	0.9888	0.8969	0.0307	**0.8372**	108.24	**1.0081**	59.11	70.80
	S2FS	**1.0039**	**0.9098**	**0.0089**	0.8103	**114.32**	0.9172	**54.69**	**66.89**
TerraSAR-X (mean value: 75.41)									
PPB	-	0.8228	0.7369	0.2693	0.3914	329.12	0.6946	**55.71**	96.41
FANS	-	0.8742	0.7864	**0.0020**	0.4379	321.77	0.8417	63.61	84.00
SAR2SAR	-	0.8262	0.7381	0.2475	0.4170	133.42	0.8865	72.96	87.18
Speckle2Void	-	0.8124	0.7323	0.1763	0.4020	160.14	0.9103	66.05	70.90
MONet	UCM	0.8491	0.7621	0.3038	0.4334	**555.34**	0.5432	58.33	92.24
	S2FS	0.8740	0.7823	0.4017	0.4366	131.86	0.7807	58.43	91.38
SAR-CAM	UCM	0.8424	0.7589	0.2765	0.4421	242.81	0.8448	66.62	87.48
	S2FS	0.8573	0.7743	0.2251	0.4418	234.30	0.8462	59.58	85.59
TransSAR	UCM	0.8425	0.7569	0.2539	0.4327	122.09	0.7331	60.01	85.83
	S2FS	0.8471	0.7648	0.2469	0.4419	209.54	0.9386	61.47	84.40
SAR-ON	UCM	0.8383	0.7569	0.2674	0.4342	515.47	0.7855	59.11	85.05
	S2FS	0.8456	0.7618	0.0748	0.4378	457.16	0.7850	59.56	83.85
Ours	UCM	0.9076	0.8330	0.0130	0.4371	233.38	**0.9854**	59.23	85.10
	S2FS	**0.9338**	**0.8629**	0.0035	**0.4454**	310.75	0.8779	58.03	**78.91**

TABLE IV
EVALUATION OF THE REAL-WORLD FARM SCENE

	set	EPD-ROA hori	EPD-ROA vert	$\beta_{\text{ratio}}\downarrow$	SQI↑	ENL↑	MOI $\rightarrow 1$	RMSE -SS	Mean
ALOS PALSAR (mean value: 36.15)									
PPB	-	0.9666	0.9356	0.0909	0.7324	151.23	0.8410	44.67	39.94
FANS	-	0.9695	0.9411	0.0306	0.7943	177.90	0.8598	39.25	41.18
SAR2SAR	-	0.9710	0.9387	0.0900	0.7348	166.63	0.8738	**28.99**	**35.43**
Speckle2Void	-	0.9659	0.9342	0.0393	0.7449	**216.09**	0.9196	51.00	39.94
MONet	UCM	0.9716	0.9414	0.0527	0.7806	112.04	0.8448	40.78	40.25
	S2FS	**0.9732**	0.9450	0.0675	0.7733	150.50	0.8338	41.07	39.89
SAR-CAM	UCM	0.9701	0.9380	0.0702	0.8020	148.13	0.9015	55.82	38.31
	S2FS	0.9689	0.9374	**0.0080**	0.8040	127.69	0.8984	46.17	38.09
TransSAR	UCM	0.9683	0.9367	0.1014	0.7847	158.27	0.8410	43.39	41.60
	S2FS	0.9699	0.9382	0.0617	0.8280	120.20	0.9027	50.27	40.16
SAR-ON	UCM	0.9685	0.9374	0.0552	0.7853	155.66	0.8749	50.90	38.59
	S2FS	0.9681	0.9369	0.0692	0.7800	166.82	0.8747	44.55	39.13
Ours	UCM	0.9695	0.9423	0.0492	0.7956	148.22	0.8527	47.04	38.26
	S2FS	0.9723	**0.9471**	0.0435	**0.8431**	211.80	**0.9243**	50.22	40.20
Sentinel-1 (mean value: 25.65)									
PPB	-	0.9363	1.4094	0.1785	0.6741	152.06	0.7710	**72.73**	33.01
FANS	-	1.0212	1.4126	0.0236	0.8148	58.76	0.8665	73.26	29.06
SAR2SAR	-	0.9237	1.4110	0.1305	0.7303	118.81	0.8315	81.48	29.96
Speckle2Void	-	0.9174	1.4065	0.1642	0.7277	163.93	1.0921	79.11	**25.19**
MONet	UCM	0.9779	1.4171	0.0995	0.8204	128.52	0.8237	73.63	29.46
	S2FS	1.0327	1.3988	0.1729	0.7989	72.83	0.9062	73.57	32.25
SAR-CAM	UCM	0.9443	1.4171	0.1317	0.7735	106.53	0.8730	79.73	30.27
	S2FS	0.9609	1.4144	0.1461	0.8014	69.78	0.8764	72.90	29.34
TransSAR	UCM	0.9235	1.4081	0.2002	0.7167	113.22	0.8154	75.94	30.78
	S2FS	0.9299	1.4171	0.1163	0.7925	153.90	0.9316	74.26	28.28
SAR-ON	UCM	0.9273	1.4003	0.1522	0.7542	171.19	0.8557	75.05	29.12
	S2FS	0.9322	1.4063	0.1242	0.7508	**186.19**	0.8436	75.01	29.70
Ours	UCM	0.9729	1.4103	0.0364	**0.8527**	115.78	0.9292	76.98	26.30
	S2FS	**0.9919**	**1.3897**	**0.0140**	0.8325	118.37	**0.9683**	73.77	27.81
TerraSAR-X (mean value: 57.99)									
PPB	-	0.9054	0.9695	0.2842	0.6248	361.37	0.8819	56.91	63.28
FANS	-	0.9109	0.9741	0.0235	0.6538	232.08	0.9600	56.92	64.81
SAR2SAR	-	0.9130	0.9749	0.2004	0.5364	175.37	0.9311	**56.01**	64.71
Speckle2Void	-	0.9070	0.9699	0.1639	0.5577	426.23	0.9525	75.49	62.37
MONet	UCM	0.9173	0.9784	0.0831	0.6432	173.35	0.9224	58.97	62.24
	S2FS	0.9176	**0.9785**	0.1155	0.6559	202.96	0.9214	58.13	66.32
SAR-CAM	UCM	0.9114	0.9747	0.0815	0.6835	300.18	0.9654	73.91	62.92
	S2FS	0.9127	0.9769	0.0387	**0.6952**	250.86	0.9594	59.94	62.56
TransSAR	UCM	0.9098	0.9722	0.0546	0.6279	445.33	0.9136	60.28	65.72
	S2FS	0.9116	0.9748	0.0383	0.6654	237.52	0.9008	60.44	64.59
SAR-ON	UCM	0.9099	0.9714	0.0858	0.6614	364.21	**0.9673**	64.19	62.73
	S2FS	0.9109	0.9744	0.0184	0.6588	252.29	0.9312	61.12	63.27
Ours	UCM	**0.9183**	0.9766	0.0088	0.6589	531.36	0.8993	66.13	62.45
	S2FS	0.9149	0.9773	**0.0045**	0.6828	**583.09**	0.8459	61.43	**61.31**

The ratio images obtained during the despeckling process of the City image from TerraSAR-X are shown in Fig. 14. Among all the methods, both FANS and the proposed method trained by S2FS maintain the fewest structures in the ratio images. Conversely, the ratio images produced by SAR2SAR, MONet, and the proposed method trained by UCM demonstrate clear structural patterns.

B. Quantitative Comparison of Real-World City and Farm Scenes

The despeckling results of SAR images from ALOS, Sentinel-1, and TerraSAR-X are evaluated. The scores are presented in Tables III and IV for City and Farm scenes, respectively. The mean values are also calculated before and after restoration. In these tables, the optimal values are highlighted in bold and the suboptimal values are underlined.

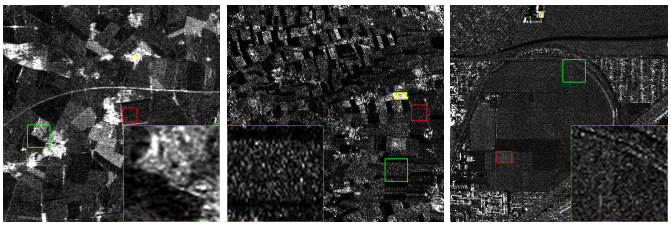
Table III presents the results of the City scene. In summary, out of the 24 comparisons, the proposed model trained with S2FS ranks first for 15 times. The proposed method trained by S2FS wins the EPD-ROA evaluations for both horizontal and vertical directions, indicating that it consistently outperforms other algorithms in terms of edge preservation and texture recovery. Our method trained by S2FS also wins the comparisons of ENL and RMSE-SS, which shows that our results have less noise in homogeneous areas and more strong scatters in heterogeneous areas.

The evaluation scores of the Farm scene are presented in Table IV. In summary, the proposed model trained with S2FS wins 10 out of the 24 comparisons. The proposed method works well for the Sentinel-1 image, which is the same to the training dataset in ground resolution.

VII. FINE-TUNE WITH MULTITEMPORAL DATA

The training dataset of this article shows its performance over different study cases, which was used in training by assuming the spatial randomness of the speckle, but real-world

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ALOS Sentinel-1 Farm 1 TerraSAR-X

Fig. 15. Real-world SAR images of the Farm scene for despeckling.

TABLE V
ENL EVALUATION FOR SENTINEL-1 SCENES WITH STACKED DATA FINE-TUNING

	City 1	City 2	City 3	Farm 1	Farm 2	Farm 3
MONet	79.72	300.43	271.66	88.75	297.58	418.35
SAR-CAM	91.04	281.56	356.85	161.06	335.26	388.75
TransSAR	76.66	335.83	272.72	249.98	332.61	404.37
SAR-ON	70.69	260.86	420.92	203.37	344.86	385.16
proposed	**323.83**	**817.17**	**1634.94**	**652.50**	**935.54**	**1576.00**

speckles are spatially correlated. To observe the difference between two types of datasets for the proposed model, a multitemporal real-world SAR dataset is built to fine-tune the network.

Domains are mismatched between real-world and simulated SAR images. Real-world SAR images are characterized by abrupt reflectance changes, geometric distortions, and high dynamics, which cannot be fully simulated by synthetic datasets. In addition, the used synthetic datasets assumed the spatial randomness of the speckle, but real-world speckles are spatially correlated.

Efforts have been made to address the domain gap issue. Lapini et al. [39] proposed a whitening method for processing a complex image acquired by an SAR system. In [23], this whitening technique was used on TerraSAR-X images for spatial decorrelation of speckles. Dalsasso et al. [40], [41] proposed to simply downsample SAR images for decorrelation with a loss of resolution. To better preserve the spatial resolution, Abergel et al. [42] proposed to correct resampled SAR images and extract bright targets. Deledalle et al. [43] achieved robustness to noise correlation by an adaptive kernel. In this article, to bridge the domain gap between synthetic and real-world SAR images, a multitemporal dataset is constructed using real-world SAR images to fine-tune the network trained on the proposed synthetic dataset.

The multitemporal real-world SAR dataset for fine-tuning is built with Sentinel-1 VV images. A total of 18 SAR images from the Sentinel-1 satellite were captured between October 4 and December 27, 2022 in Strasbourg, France. Each SAR image was cropped into 1600 patches with a size of 256 × 256 and a step size of 128. Among the 1600 image sequences, the sequences with large variations were removed according to [44]. Eventually, 455 image sequences were retained to construct the multitemporal dataset. Each image sequence provides a pair of images for supervised learning, where the input is an SAR image from November 05, 2022, and the corresponding ground truth is obtained by averaging the image sequence. In the fine-tuning experiment, this dataset is used to fine-tune the model that has been trained with S2FS, where 400 images are used for training and 55 images are for validation.

In the fine-tuning stage, the learning rate was reduced to one-tenth of the original values. The competing methods were trained over ten epochs. As for our model, the number of total time steps was extended to 1150 and only 1 epoch was trained. After the log-Yeo–Johnson transformation for the stacked multitemporal dataset, the remaining weak speckle noise in the stacked image becomes the log-Yeo–Johnson transformed Gaussian-like noise that was mixed in the forward diffusion. Therefore, we hope to extend the time steps in fine-tuning to decrease the noise given the same time step, such that the noise in stacked data contributes to the whole noise approaching the noise in the first training.

Since the stacked dataset was constructed using Sentinel-1 images, fine-tuning was tested only for Sentinel-1. In addition to the Sentinel-1 images used in the real-world test, two groups of newly gathered Sentinel-1 images are included for the fine-tuning test which are shown in Fig. 17. The ENL comparison is shown in Table V. The ENL region size of City 2 is 15 × 20, and the ENL region size of City 3 is 40 × 15. The ENL region size of Farm 2 is 20 × 25, and the ENL region size of Farm 3 is 40 × 40, as shown in Fig. 16. It can be concluded from Fig. 16 and Table V that fine-tuning has a positive impact for diffusion models, as it can significantly remove the speckles while retaining a good structure.

VIII. DISCUSSION

Three experiments are made in this section. The first experiment studies the impact of preprocessing methods. The second experiment analyzes the effectiveness of the hybrid training for diffusion. The last experiment explores alternative sampling methods for the backward diffusion process. The importance of the training set is also discussed.

A. Ablation Study—Preprocessing and Postprocessing

This ablation study clarifies the impact of the log-Yeo–Johnson transformations in the diffusion model. To this aim, the diffusion model was trained using the 1-look S2FS dataset in three different scenarios. In the first scenario, Gaussian noise is simulated and gradually added to the reference image to form noisy images. In the second scenario, Gaussian noise is still simulated but added to the log-transformed reference images to form noisy images. Here, it is assumed that the distribution of the logarithm of the Gamma noise matches the Gaussian distribution [19]. While in the third scenario, Gamma noise is simulated and applied both the logarithm and Yeo–Johnson transformations to adjust its distribution toward a Gaussian distribution. This log-Yeo–Johnson transformed noise is then added to the log-Yeo–Johnson transformed reference images to form noisy images. The diffusion model is trained in these three scenarios, and the results are reported in Table VI. Table VI shows that no difference exist between these methods. Therefore, it is unnecessary to preprocess when only synthetic datasets are used. This conclusion is

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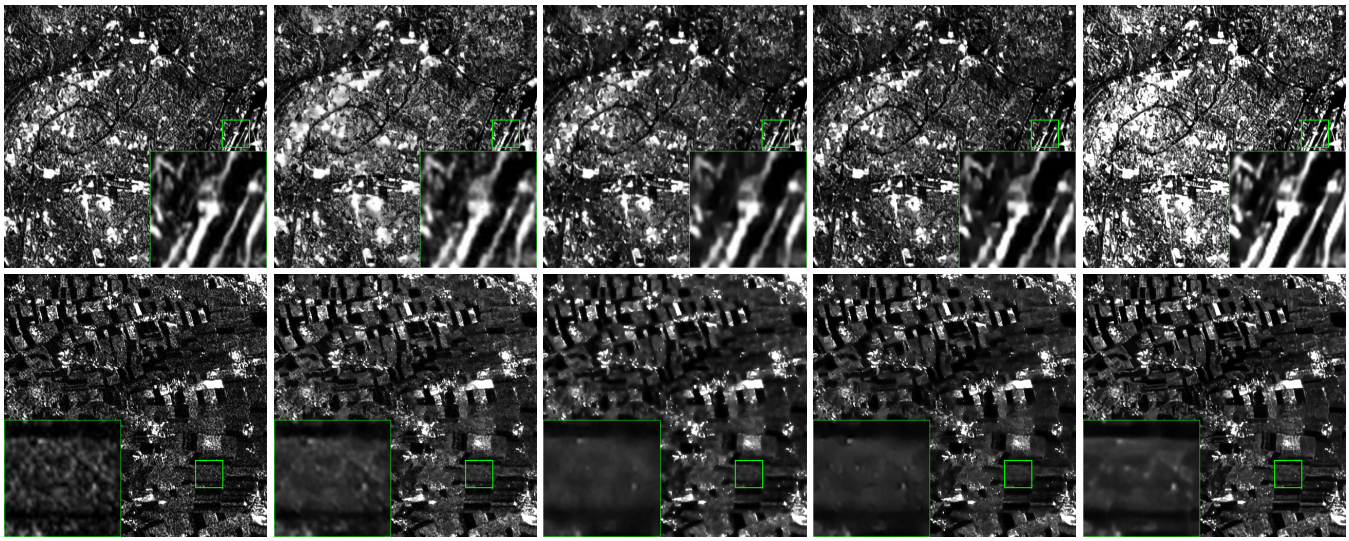


Fig. 16. Fine-tuning real-world Sentinel-1's City and Farm scenes with stacked multitemporal data.

TABLE VI
IMPACT OF VARIOUS PREPROCESSING FOR SYNTHETIC SCENES

Log	Yeo-Johnson	Farm		City		Forest		Port	
		SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
×	×	0.8606	30.30	0.7348	22.33	0.8970	36.56	0.7887	23.36
✓	×	0.8622	30.00	0.7357	22.18	0.9064	36.61	0.7952	23.41
✓	✓	0.8701	30.05	0.7382	22.33	0.9050	36.57	0.7928	23.38

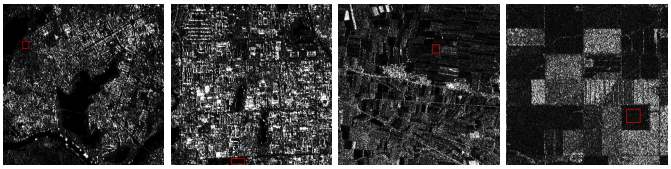


Fig. 17. Real-world Sentinel-1 images of the City and Farm scenes for fine-tuning.

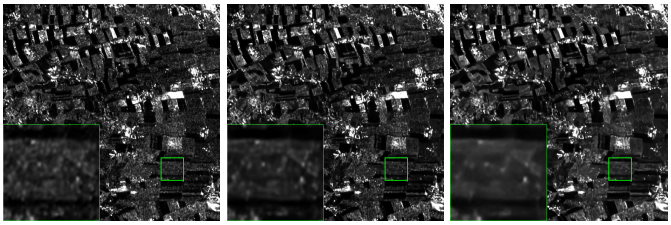


Fig. 18. Finetuning with various preprocessing solutions.

consistent to our explanation to the nature of learning clean data distribution with the diffusion model.

The differences between the three methods are amplified in the fine-tuning stage. Fine-tuning was performed for the three combinations, and the results are presented in Fig. 18. It is clearly observed that the logarithmic transformation can alleviate the speckles when the first two images are compared. The Yeo–Johnson transformation is very effective as the noise in the third image is significantly smoothed. It is noteworthy that the strong scatters in the third image is preserved well. Therefore, our preprocessing solution finds the tradeoff between smoothing speckles and maintaining strong scatters.

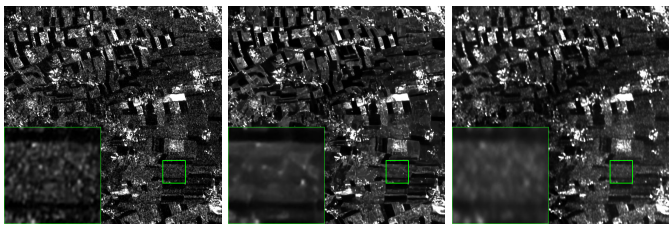


Fig. 19. Despeckling real-world Sentinel-1 images of the Farm scene with the proposed method and three training methods (synthetic, multitemporal fine-tuning, and hybrid).

B. Ablation Study—Training Methods

Vitale et al. [44] proposed a hybrid training by combining multitemporal and synthetic training methods. When hybrid training is used for neural networks, the target data is an average image from a multitemporal SAR dataset, which is processed with (1) and (2) to simulate the real SAR image. In other words, averaged SAR images are prepared for synthetic training instead of clean optical images. This hybrid training method was tested in this section to verify its effectiveness for our model.

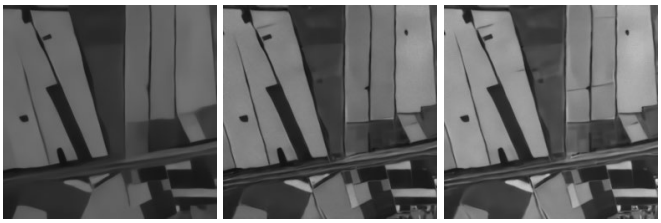
Three datasets from the Sentinel-1 VV band were used for the hybrid training. The first dataset is the multitemporal dataset used in fine-tuning. The second dataset was taken from Erfurt of Germany between September 5, 2022 and December 27, 2022. The third dataset was collected from Padua of Italy between September 6, 2022 and December 30, 2022. Patches were extracted, selected, and averaged from these datasets. Finally, a total of 1716 pairs of 256 × 256-sized SAR image patches were extracted for training. The parameter settings and loss functions are consistent with those in the synthetic training.

The Farm results of hybrid training for the proposed method are listed in Fig. 19, which are companied with the results of synthetic and multitemporal fine-tuning training. By comparing the images in Fig. 15, the hybrid training is also effective

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TABLE VII
TESTS WITH ALTERNATIVE BACKWARD DIFFUSION METHODS

	condition	Farm		City		Forest		Port	
		SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR
log [19]	×	0.8001	20.614	0.4375	13.786	0.8388	33.235	0.5859	16.917
log	✓	0.8745	28.047	0.7361	19.551	0.9089	36.582	0.7924	22.430
log-Yeo-Johnson	✓	0.8740	27.833	0.7428	20.419	0.9047	36.253	0.7928	22.543



log without condition log log-Yeo-Johnson

Fig. 20. Despeckling on the simulated 1-look S2FS dataset.

in removing SAR speckles. However, the results of the hybrid training are more noisy, and the structures are less clear. The reason is partially explained by the average of multitemporal data which may blur the key features in noisy SAR images and prevent the proposed diffusion model to accurately learn high-frequency details.

C. Alternative Sampling Technique

In addition to the traditional test that starts from a purely noisy image for diffusion models, a test method specifically applicable to SAR despeckling was proposed by Xiao et al. [19]. The alternative method employs a logarithmized SAR image as the noisy image x_t in the reverse diffusion process. Instead of starting from T, the time step t in the reverse process starts from a small number and gradually iterates to 0. This generation method can also be used for the method proposed in this article.

Three methods were designed for testing. The first method trained the diffusion model on the S2F2 synthetic dataset using the network structure of this article plus a logarithmic transformation, but the conditional branch was removed. The second method is similar to the first one, and the conditional image is harnessed for reference. The third method is similar to the second one except for the logarithmic transformation replacing by the log-Yeo–Johnson transformation for preprocessing. The three tests were performed on the S2F2 synthetic dataset. The number of total time steps T in the backward diffusion was adjusted to searching for suitable values.

The experimental results show that better performance can be obtained by choosing the number of backward diffusion steps T deliberately. The first method (without conditional images) removes noise effectively when T is 60. The second method using only the logarithmic transformation removes the vast majority of the noise with about 100 steps. The conditional diffusion model using the log-Yeo–Johnson transform removes almost all of the noise with about 70 steps. The experimental results are evaluated and presented in Table VII. The partly manifested results are presented in Fig. 20.

The results in Table VII and Fig. 20 show that better performance is expected when the conditional branch is introduced. When the conditional image is not used, the result is oversmoothed and the score is low. Although the speckle is effectively removed, noticeable features in the result are removed as well. When the condition module is used, the despeckling effect of only performing logarithmic transformation is similar to that of the log-Yeo–Johnson preprocessing. The former requires 100 steps to approaching the goal while the latter needs 70 steps. It is therefore concluded that the proposed method adapts to the alternative generation technique well when suitable backward steps are given.

D. Importance of Training Data

The importance of the dataset in diffusion modeling is further confirmed through experimentation. Two training datasets with a significant difference in resolution were used for the study. As evidenced by Figs. 10–14, the significant difference in results between the models trained by UCM and S2FS validates the effectiveness of the newly designed dataset in training the diffusion model for satellite SAR image despeckling. Since diffusion models aim to approximate the distribution of real data, using a similar scale can help in learning a more accurate statistical pattern.

IX. CONCLUSION

In this article, we present DiffusionSAR, a novel method based on the conditional diffusion model for SAR image despeckling. The proposed method leverages a conditional probabilistic diffusion model to explicitly learn the specific distributions within the training data. It employs an iterative noise modeling process, which combines forward addition and backward denoising, to effectively despeckle SAR images while recovering essential structural information. The log-Yeo–Johnson transformation is used to convert the multiplicative gamma noise into additive Gaussian noise approaching the distribution in diffusion process. A new synthetic dataset is designed with Sentinel-2 images to train the model. A new multitemporal dataset is also designed for the fine-tuning test. The proposed method is compared with eight methods using synthetic datasets and real-world SAR images for performance evaluation. Additional experiments, including fine-tuning training, hybrid training, and alternative sampling methods, uncover the potential of the proposed method in achieving better performance.

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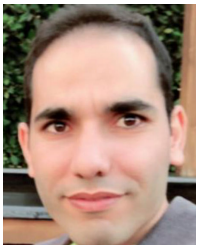
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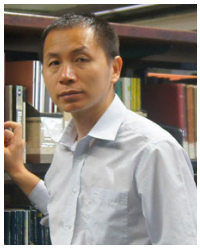
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